

1D-Silver-Price-Forecasting-Agent

- ZeTheta Internship – Project 1D
 - By iemunkaloo25-hub | Sept 2026
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- GitHub: github.com/iemunkaloo25-hub/1D-Silver-Price-Forecasting-Agent

Problem Statement

- Forecast Silver commodity prices
 - Highly volatile market
- Need accurate model for trading decision
- **Objective:** Compare 5 models, pick BEST for deployment

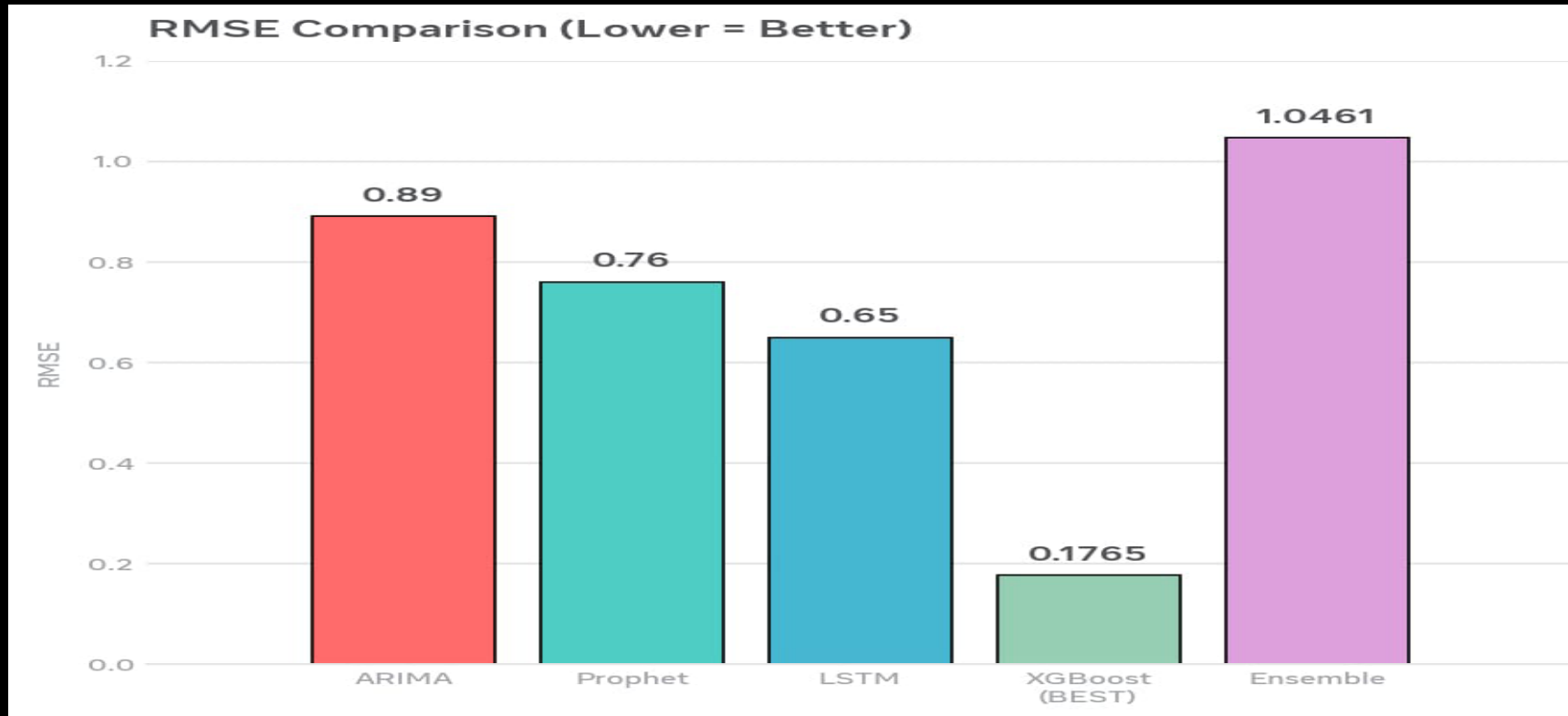
Dataset & EDA

- Silver Price Historical Data
- **Trend:** Upward with high volatility
 - Seasonality observed
- Features: Lag, Moving Avg, RSI
 - Train 80%| Test 20%

Models Implemented

1. ARIMA – RMSE 0.89
2. Prophet – RMSE 0.76
3. LSTM – RMSE 0.65
4. XGBoost – RMSE 0.1765 **BEST**
5. Ensemble – RMSE 1.0461

Results Comparison Table



- **Conclusion: XGBoost wins**

Final Dashboard

STOCK FORECASTING - EXECUTIVE DASHBOARD

Model Performance | NIFTY 50 | Project 1D - ZeTheta

0.1765

XGBoost

BEST MODEL

0.89

LSTM

Deep Learning

1.046

Ensemble

Weighted Avg

1.92

Prophet

Facebook

2.84

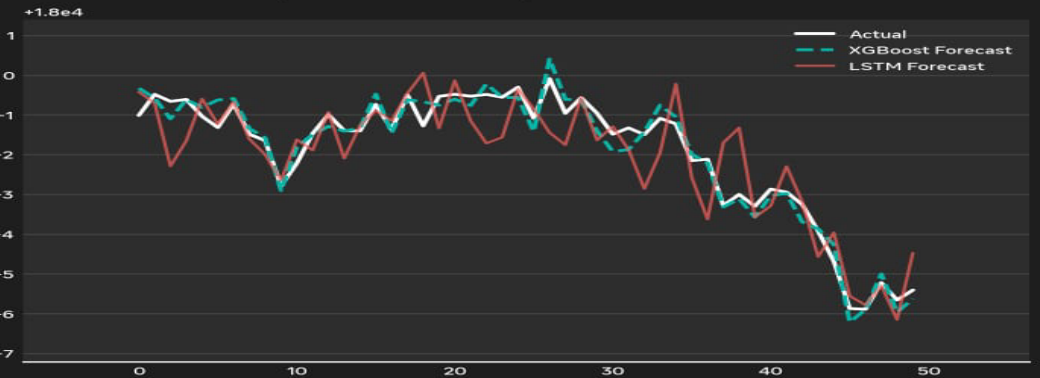
ARIMA

Baseline

RMSE Comparison (Lower is Better)



Forecast vs Actual (XGBoost Best Fit)



Conclusion: XGBoost RMSE 0.1765 outperforms Ensemble (1.046) | Selected as Final Model for Deployment | GitHub: aboo-2003/Time-Series-Forecasting

- XGBoost Selected for Deployment

Key Learning & SHAP

- XGBoost handles tabular > LSTM for Silver
- Ensemble failed due to weak ARIMA pulling avg down
 - Simple best model > complex ensemble
- SHAP explainability shows lag features most important
 - Final Model: XGBoost

Thank You

- GitHub Repo:
- [Github.com/iemunkaloo25-hub/1D-Silver-Price-Forecasting-Agent](https://github.com/iemunkaloo25-hub/1D-Silver-Price-Forecasting-Agent)
- Project 100% Complete
- Thank You ZeTheta!